MECH 325 Machine Elements

Gear Trains

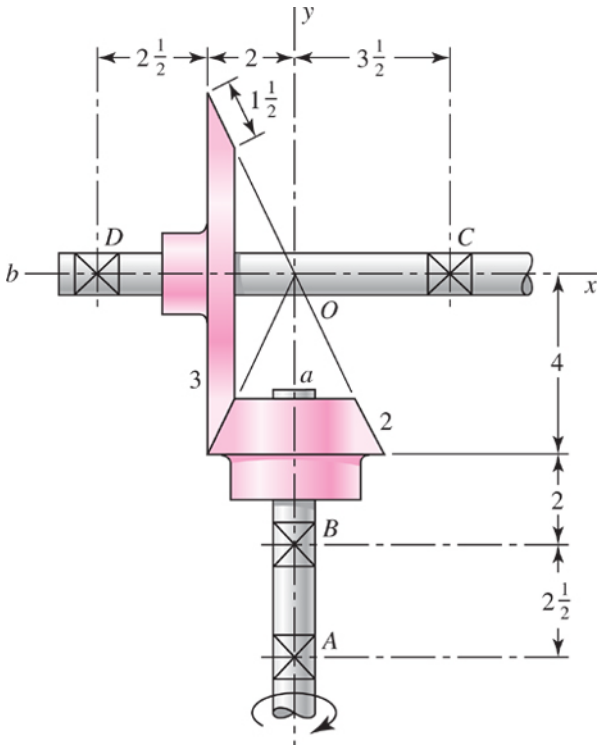
Tutorial 2 – Gear Kinematics and Gear Force Analysis

1. Modified Shigley, Problem 13-49 (9th and 10th Ed. 13-43, 8th Ed., 13-29). **(To be demonstrated by the TA.)**

The figure shows a 16T 20° straight bevel pinion driving a 32T gear, and the bearing centerlines. Pinion shaft *a* receives 2.5 hp at 240 rev/min. Assume the gear contact forces occur at the midpoints of the teeth.

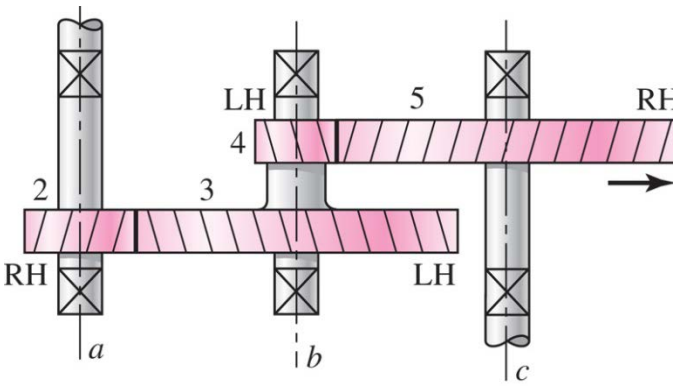
a. Determine the contact force vector.

b. Draw the free body diagram for shaft *a* including the bearing reactions at *A* and *B*. Assume bearing *A* is to take both radial and thrust loads.



2. **Shigley, Problem 13-53** (10th Ed. 13-47, 8th Ed., 13-33). **(To be done by students.)**

A gear train is composed of four helical gears with the three shaft axes in a single plane, as shown in the figure. The gears have a normal pressure angle of 20° and a 30° helix angle. Gear 2 is the driver, and is rotating CCW when viewed from the top. Shaft *b* is an idler and the transmitted load acting on gear 3 is 500 lbf. The gears on shaft *b* both have a normal diametral pitch of 7 teeth/in and have 54 and 14 teeth, respectively. Find the forces exerted by gears 3 and 4 on shaft *b* (don't worry about determining moments)



3. **Note: If time permits. Shigley, Problem 13-25** (8th Ed., 13-18). **(To be done by students)**
   Note: A model differential video may be shown for the tutorial.

The tooth numbers for the automotive rear differential are $N_2 = 16$, $N_3 = 48$, $N_4 = 14$, $N_5 = N_6 = 20$. The drive shaft is turning at 900rpm. The vehicle is a rear wheel drive car.

a) What are the wheel speeds if the car is travelling in a straight line on a good road surface?

b) Suppose the rear right wheel is jacked up and the left wheel is resting on a good road surface, what is the speed of the right wheel?

c) If the right rear wheel is now resting on an icy surface and the left wheel is on dry pavement, what will happen if you attempt to drive away? Can you think of a way to get going?

